

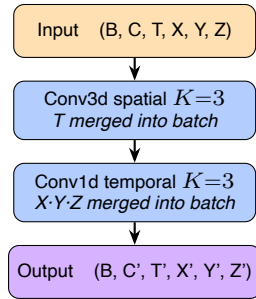
Patchify architecture — PatchEmbed3DPlus1D

2026-06-14

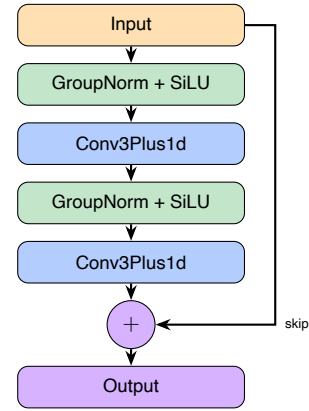
Hierarchical 3D+1D patchify encoder, inspired by the MovieGen TAE TemporalEncoder (tae.py:1052).

Building blocks

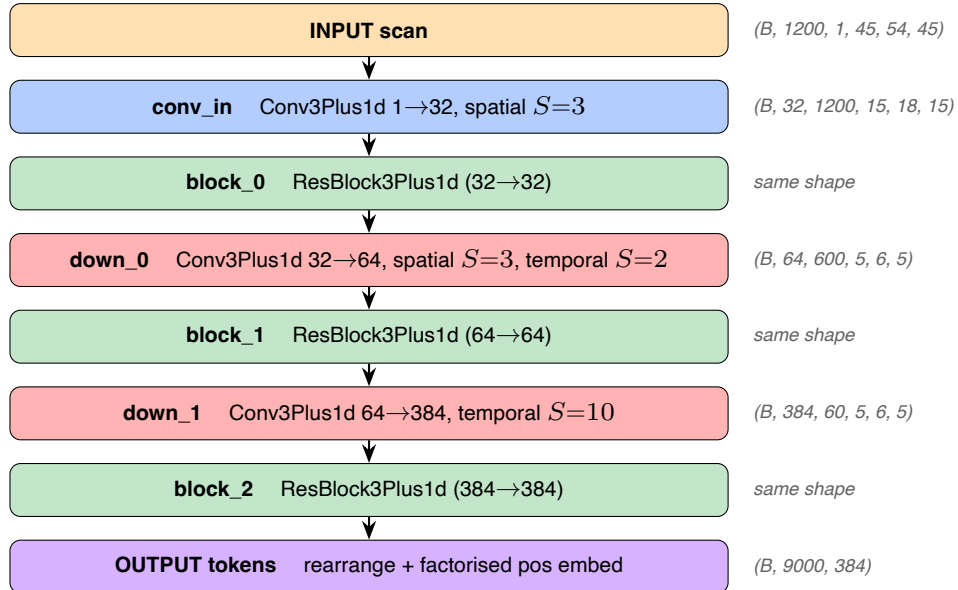
Conv3Plus1d — separable substitute for the missing PyTorch Conv4d. Splits a 4D conv into a 3D spatial conv (per frame) followed by a 1D temporal conv (per voxel-position).



ResBlock3Plus1d — residual block of 2 Conv3Plus1d with GroupNorm + SiLU pre-activation, output added to a skip connection. Same shape in / out.



Full encoder (HCP-YA setup, T = 1200, temporal_kernel = 20)



Parameter breakdown

Layer	What it contains	Params
conv_in	1 Conv3Plus1d (1{→}32)	4 000
block_0	2 Conv3Plus1d (32{→}32) + 2 GroupNorm	61 696
down_0	1 Conv3Plus1d (32{→}64)	67 712
block_1	2 Conv3Plus1d (64{→}64) + 2 GroupNorm	246 272
down_1	1 Conv3Plus1d (64{→}384)	2 138 880
block_2	2 Conv3Plus1d (384{→}384) + 2 GroupNorm	8 850 432
PositionEmbedding3D	pos_temporal + pos_spatial + pos_cls	81 024
TOTAL		11 450 016

Token grid output : $T_{\text{eff}} = 60$, $N_{\text{spatial}} = 5 \cdot 6 \cdot 5 = 150$, total **9 000 tokens** of dim 384.

Detailed parameter calculation per layer

Each Conv3Plus1d has 2 internal modules : Conv3d(in_c, out_c, K_s) (spatial) + Conv1d(out_c, out_c, K_t) (temporal). Both convs include a bias term.

Conv3d params = $\text{in}_c \cdot \text{out}_c \cdot K_s^3 + \text{out}_c$ **Conv1d** params = $\text{out}_c \cdot \text{out}_c \cdot K_t + \text{out}_c$ **GroupNorm** params = $2 \cdot C$ (scale + shift, both of length C)

1. conv_in = Conv3Plus1d(1 → 32, $K_s=3$, $K_t=3$)

spatial	Conv3d(1, 32, k=3)	:	$1 \times 32 \times 3^3 + 32$	=	896
temporal	Conv1d(32, 32, k=3)	:	$32 \times 32 \times 3 + 32$	=	3 104
					4 000

2. block_0 = ResBlock3Plus1d(32) = 2x Conv3Plus1d(32 → 32) + 2x GroupNorm(8, 32)

GroupNorm × 2	:	$2 \times (2 \times 32)$	=	128
Conv3Plus1d × 2 :				
spatial each	:	$32 \times 32 \times 3^3 + 32$	=	27 680 each
temporal each	:	$32 \times 32 \times 3 + 32$	=	3 104 each
total one	:	$27 680 + 3 104$	=	30 784
× 2	:		=	61 568
				61 696

3. down_0 = Conv3Plus1d(32 → 64, $K_s=3$, $K_t=3$)

spatial	Conv3d(32, 64, k=3)	:	$32 \times 64 \times 3^3 + 64$	=	55 360
temporal	Conv1d(64, 64, k=3)	:	$64 \times 64 \times 3 + 64$	=	12 352
					67 712

4. block_1 = ResBlock3Plus1d(64) = 2x Conv3Plus1d(64 → 64) + 2x GroupNorm(8, 64)

GroupNorm × 2	:	$2 \times (2 \times 64)$	=	256
Conv3Plus1d × 2 :				
spatial each	:	$64 \times 64 \times 3^3 + 64$	=	110 656 each
temporal each	:	$64 \times 64 \times 3 + 64$	=	12 352 each
total one	:	$110 656 + 12 352$	=	123 008
× 2	:		=	246 016
				246 272

5. down_1 = Conv3Plus1d(64 → 384, $K_s=3$, $K_t=10$)

spatial Conv3d(64, 384, k=3) : $64 \times 384 \times 3^3 + 384 = 663\,936$
temporal Conv1d(384,384, k=10) : $384 \times 384 \times 10 + 384 = 1\,474\,944$

2 138 880

6. block_2 = ResBlock3Plus1d(384) = 2× Conv3Plus1d(384 → 384) + 2× GroupNorm(8, 384)

The dominant layer — 77% of the total.

GroupNorm × 2 : $2 \times (2 \times 384) = 1\,536$
Conv3Plus1d × 2 :
 spatial each : $384 \times 384 \times 3^3 + 384 = 3\,981\,696$ each
 temporal each : $384 \times 384 \times 3 + 384 = 442\,752$ each
 total one : $3\,981\,696 + 442\,752 = 4\,424\,448$
 × 2 : $= 8\,848\,896$

8 850 432

7. PositionEmbedding3D (HCP-YA : $T_{\text{eff}}=60$, $N_{\text{spatial}}=150$, dim =384)

pos_temporal : $60 \times 384 = 23\,040$
pos_spatial : $150 \times 384 = 57\,600$
pos_cls : $1 \times 384 = 384$

81 024

Grand total: $4\,000 + 61\,696 + 67\,712 + 246\,272 + 2\,138\,880 + 8\,850\,432 + 81\,024 = 11\,450\,016$ params.

Differences vs MovieGen TAE TemporalEncoder

Aspect	MovieGen TAE	Ours (PatchEmbed3DPlus1D)
Domain	Video (T, H, W)	fMRI volumes (T, X, Y, Z)
Building block	Conv2Plus1d (2D spatial + 1D temporal)	Conv3Plus1d (3D spatial + 1D temporal)
Hierarchical levels	4 (default ch_mult=(1,2,4,8))	3 (channels 32 / 64 / 384)
ResBlocks per level	2	1
Attention in encoder	Yes (at specified resolutions)	None
Middle bottleneck section	2 ResBlocks + 1 AttnBlock	None
Output	Latent tensor z (for decoder)	Token sequence (B, $T_{\text{eff}} \cdot N_{\text{spatial}}$, 384) for ViT